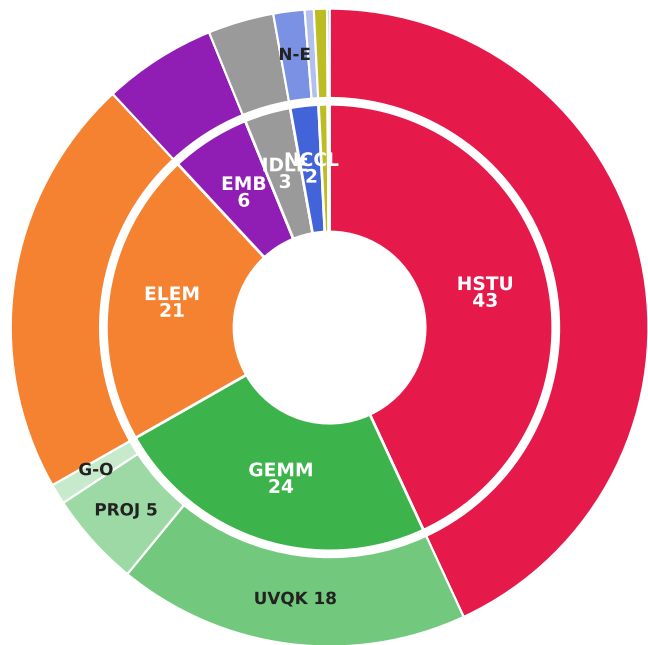
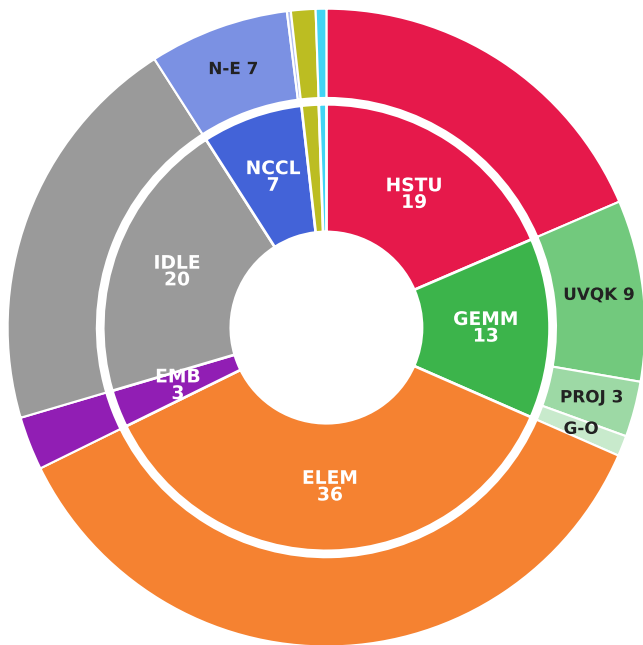


HSTU GPU-time sunburst — NON-JAGGED (matched config), exposed accounting, fastest step (upstream \$2.2 method)
inner ring = grouping · outer ring = leaves · % of the step

Upstream H100 · step 162 · D256 (nj)



Our GB300 · step 153 · D128 (nj)



Legend:

- HSTU = hstu fwd/bwd
- UVQK = gemm / uvqk
- PROJ = gemm / projection
- G-O = gemm / others
- GEMM = gemm (ours: not sub-split)
- ELEM = elementwise
- EMB = embedding op
- NCCL = nccl (exposed)
- IDLE = GPU idle
- OVL = overlapped
- OTH = others